

AhangTech: Full-Stack Web Platform Specification

Domain: ConTech (Construction Technology) | Architecture: Decoupled API + Web Frontend

1. Executive Summary & Architecture Shift

AhangTech is transitioning from a terminal-based CLI tool into a production-grade, multi-user ConTech platform. The core mission remains central: eliminating data fragmentation across Asset Management, Daily Site Reporting, and Material Estimation. The platform serves site managers, field supervisors, procurement officers, and operational contractors via an intuitive visual dashboard.

2. Selected Technology Stack

Component	Selection	Rationale
Backend API	FastAPI (Python 3.10+)	Enables direct reuse of core domain logic with automatic OpenAPI/Swagger documentation generation.
Database / ORM	SQLAlchemy (SQLite Dev / Postgres Prod)	Replaces local CSV/JSON with transactional safety, relational foreign keys, and migration readiness.
Frontend Dashboard	React (Vite) + Tailwind CSS (Option A) or Jinja2 + HTMX (Option B)	Provides reactive dashboards for real-time asset tracking and mobile-responsive on-site forms.
Validation	Pydantic v2	Strict request/response type validation and boundary sanitization.

3. Directory Structure

```
AhangTech/
|-- README.md           # Project overview & full-stack setup instructions
|-- requirements.txt     # Python backend dependencies (fastapi, uvicorn, sqlalchemy)
|-- backend/
|   |-- main.py          # FastAPI entrypoint, middleware & route mounting
|   |-- database.py       # DB engine, Base class, and session factory
|   |-- models.py         # SQLAlchemy ORM models (Asset, User, Report, Material)
|   |-- schemas.py        # Pydantic data schemas
|   |-- routers/          # Modular REST endpoints
|       |-- assets_router.py # CRUD and state transitions for equipment
|       |-- reporter_router.py # Daily site logs & external OpenWeather API integration
|       |-- estimator_router.py # Cost and dimension estimation endpoints
|-- frontend/            # Web Client Interface (components, pages, services)
```

4. Sequential Implementation Roadmap

- Phase 1: Asset Manager Full-Stack MVP (Current Focus)** — REST endpoints for /api/assets (GET, POST, PUT, DELETE). Enforce strict checkout logic (e.g., return HTTP 400 if checking out an item tagged 'Under Repair'). Frontend inventory table with check-out modals.
- Phase 2: The Reporter Module** — Daily log schema and OpenWeather API integration to automatically capture atmospheric parameters on submit. Historical log feed UI.
- Phase 3: The Estimator Module** — Endpoint calculation taking dimensions, unit costs, and material waste factors to produce precise procurement figures.